

INTERNSHIP PROJECT REPORT

Exploratory Data Analysis of the Zomato Restaurant Dataset

A Data Science & Analytics Internship Submission

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Role	AI / ML & Data Science Intern
Tool Used	Jupyter Notebook (Python)
Dataset	Zomato Restaurant Dataset (9,551 records, 21 columns)
Libraries Used	pandas, numpy, matplotlib, seaborn, folium
Report Date	29 June 2026

Submitted as part of the Data Science & Analytics Internship Program

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1. Introduction

This report documents the exploratory data analysis (EDA) carried out on the Zomato Restaurant Dataset as part of a Data Science & Analytics internship assignment. The objective of the project was to apply core data analysis skills — data cleaning, aggregation, statistical summarisation, and visualisation — to a real-world, semi-structured restaurant dataset and derive actionable business insights.

The analysis was carried out in a Jupyter Notebook using Python, and was structured into three progressive difficulty levels (Level 1, Level 2, and Level 3), each containing a series of guided tasks. This structure mirrors common industry-style data analyst internship assignments and is designed to build proficiency in data wrangling, descriptive statistics, correlation analysis, and visual storytelling with data.

1.1 Objectives

- Load, inspect, and clean a real-world restaurant dataset.
- Identify the most common cuisines and their market share.
- Analyse restaurant distribution and ratings across cities.
- Study pricing patterns and their relationship with restaurant features.
- Examine the impact of online delivery availability on customer ratings.
- Explore the geographic spread of restaurants using coordinate data.
- Identify restaurant chains and evaluate their performance and popularity.

1.2 Tools and Technologies

Component	Detail
Environment	Jupyter Notebook (JupyterLab interface)
Language	Python 3.11
Core Libraries	pandas, numpy
Visualization	matplotlib, seaborn
Geospatial Mapping	folium
Dataset Size	9,551 rows × 21 columns

2. Dataset Overview

The dataset used for this project is the well-known Zomato restaurant listings dataset, which contains detailed information on restaurants across multiple countries, with a strong concentration of records from India and the Philippines. The dataset was loaded directly from a CSV file using pandas.

2.1 Loading the Data

The dataset was loaded and libraries imported using the following setup:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv("Dataset.csv")
```

2.2 Structure and Data Types

The dataset comprises 9,551 entries across 21 columns, combining identifiers, categorical descriptors, geographic coordinates, and numeric rating/voting metrics. Key columns include:

Column	Description	Data Type
Restaurant ID / Name	Unique identifier and name	int64 / str
City / Address / Locality	Geographic descriptors	str
Longitude / Latitude	Geographic coordinates	float64
Cuisines	Comma-separated cuisine list	str
Average Cost for two	Approx. cost for two diners	int64
Price range	Price tier (1–4)	int64
Has Table booking / Has Online delivery	Yes/No service flags	str
Aggregate rating	Average customer rating (0–5)	float64
Votes	Number of customer votes	int64

2.3 Data Quality Check

A missing-value check (`df.isnull().sum()`) confirmed that the dataset is almost entirely complete: only the Cuisines column contains a small number of missing entries (9 out of 9,551 rows), which were excluded using `.dropna()` during cuisine-related calculations. All other fields are fully populated, indicating a high-quality, analysis-ready dataset that required minimal cleaning.

3. Level 1 Analysis

Level 1 focuses on foundational descriptive analysis: cuisine popularity, city-level distribution, price range patterns, and the effect of online delivery on ratings.

3.1 Task 1 — Top Cuisines

The Cuisines column was split on commas and exploded into individual cuisine entries to compute frequency counts across the full dataset.

Rank	Cuisine	Restaurant Count	% of Total Restaurants
1	North Indian	3,960	41.46%
2	Chinese	2,735	28.61%
3	Fast Food	1,986	20.80%

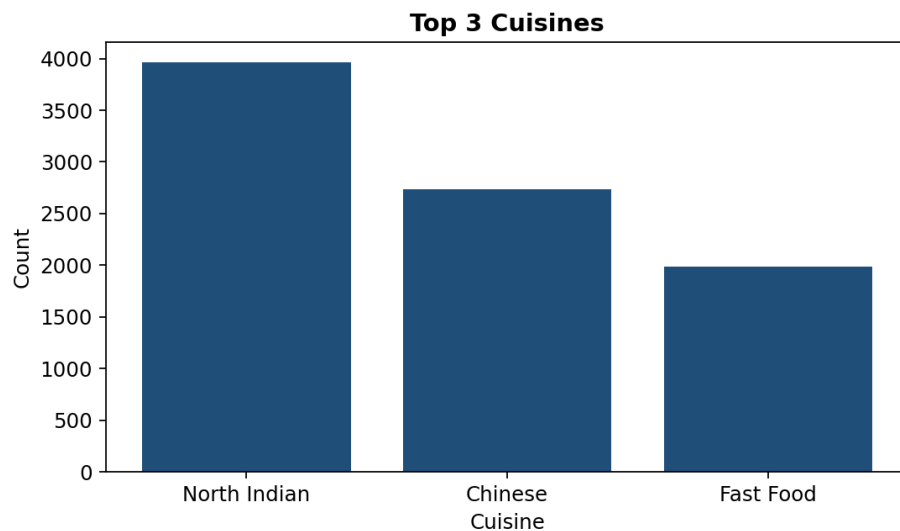


Figure 1: Top 3 cuisines by restaurant count

Insight: North Indian cuisine dominates the dataset, appearing in over 41% of all restaurant listings — more than the next two most common cuisines combined relative to their individual shares. This reflects the heavy representation of Indian cities (particularly New Delhi and the National Capital Region) in the dataset.

3.2 Task 2 — City Analysis

City-level aggregation was performed using `value_counts()` for restaurant volume and `groupby().mean()` for average ratings.

City with Highest Number of Restaurants: New Delhi (5,473 restaurants)

Highest Average-Rated City: Inner City (average rating 4.9)

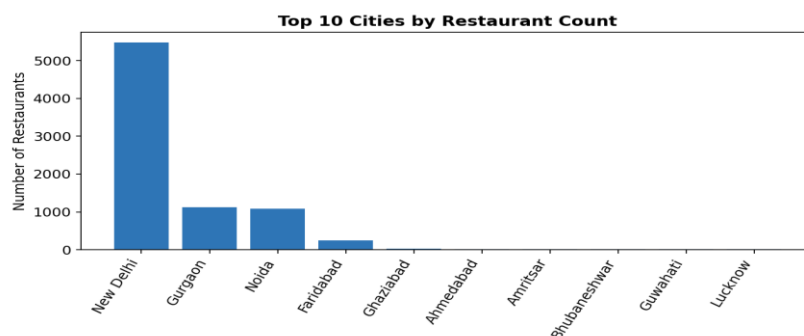


Figure 2: Top 10 cities by restaurant count

Insight: New Delhi overwhelmingly dominates restaurant volume, consistent with the dataset's strong NCR (National Capital Region) bias — New Delhi, Gurgaon, Noida, and Faridabad together account for the vast majority of Indian listings. However, the highest-rated cities (Inner City, Quezon City, Makati City) are smaller markets, suggesting that restaurant quality perception is not necessarily correlated with market size.

3.3 Task 3 — Price Range Distribution

Price Range	Restaurant Count	Percentage
1 (Budget)	4,444	46.53%
2 (Moderate)	3,113	32.59%
3 (Expensive)	1,408	14.74%
4 (Premium)	586	6.14%

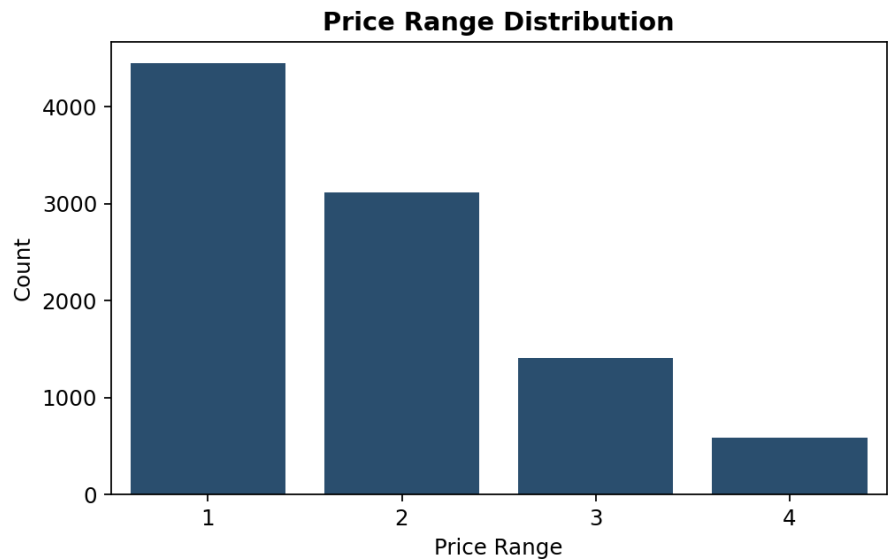


Figure 3: Distribution of restaurants across price ranges

Insight: Nearly half of all restaurants (46.5%) fall into the most budget-friendly price tier, and almost 80% fall within price ranges 1–2. This indicates that the dataset — and the markets it represents — is heavily weighted toward affordable, everyday dining rather than premium establishments.

3.4 Task 4 — Online Delivery Impact

Restaurants Offering Online Delivery: 25.66% Yes | 74.34% No

Has Online Delivery	Average Aggregate Rating
No	2.47
Yes	3.25

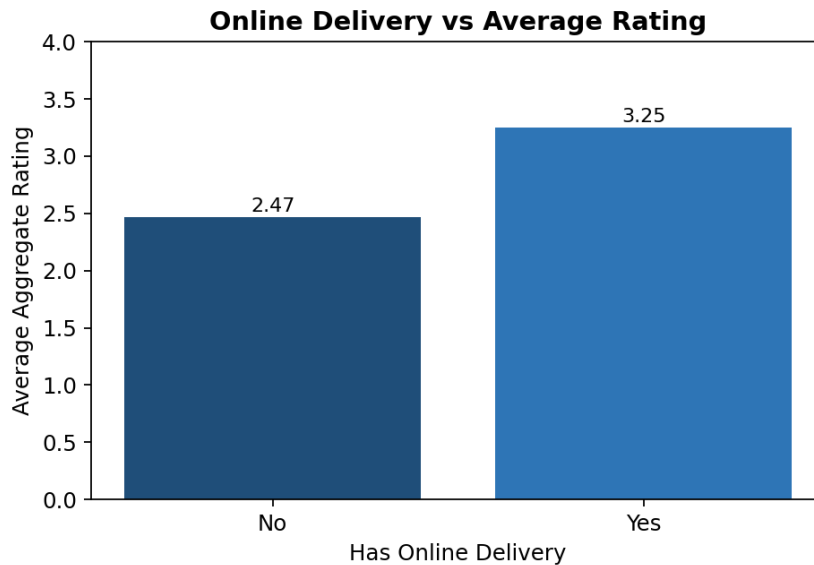


Figure 4: Average rating comparison — restaurants with vs. without online delivery

Insight: Restaurants offering online delivery have a noticeably higher average rating (3.25) than those that do not (2.47). This roughly 0.78-point gap suggests that online delivery availability is associated with stronger overall customer satisfaction — though this is a correlation, not necessarily a causal relationship, since delivery-enabled restaurants may also be more established or service-oriented overall.

4. Level 2 Analysis

Level 2 extends the analysis to ratings behaviour, cuisine combinations, geographic distribution, and multi-location restaurant chains — introducing more advanced aggregation and visualization techniques.

4.1 Task 1 — Restaurant Ratings

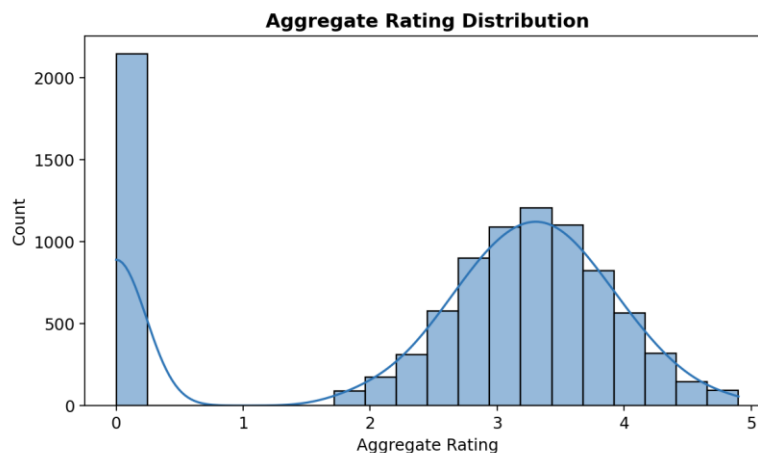


Figure 5: Distribution of aggregate ratings across all restaurants

Most Common Rating Range: 3–4 (4,388 restaurants), followed by 2–3 (1,891) and 4–5 (1,114).

Average Votes per Restaurant: 156.91

Insight: The rating distribution is bimodal — a large spike at 0 represents restaurants with no rating yet (newly listed or insufficiently reviewed), while the remaining ratings follow a roughly normal distribution centred around 3.0–4.0. Excluding unrated restaurants, the typical restaurant in this dataset is rated as “Good” (3–4 range) rather than exceptional or poor.

4.2 Task 2 — Cuisine Combinations

Beyond single cuisines, the most frequent multi-cuisine combinations were also analysed:

Cuisine Combination	Restaurant Count
North Indian	936
North Indian, Chinese	511
Chinese	354
Fast Food	354
North Indian, Mughlai	334

The highest-rated cuisine combinations (average rating 4.9) included niche pairings such as Burger/Bar Food/Steak, American/Burger/Grill, and American/Caribbean/Seafood — specialised, smaller-scale menus that tend to attract very loyal, satisfied customer bases.

Insight: While North Indian and Chinese cuisines dominate by volume, the highest customer satisfaction is associated with niche, tightly-focused cuisine combinations rather than broad, generalist menus — suggesting a potential trade-off between market reach and perceived quality.

4.3 Task 3 — Geographic Analysis

Restaurant locations were plotted using their longitude and latitude coordinates, and an interactive map was additionally generated using the folium library (exported as restaurant_map.html with markers for the first 500 restaurants).

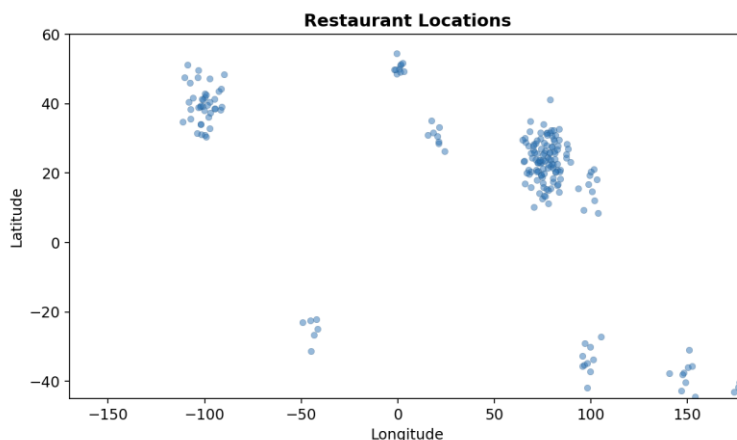


Figure 6: Geographic distribution of restaurants by longitude and latitude

Descriptive statistics of the coordinates confirmed a mean latitude of 25.85° and mean longitude of 64.13°, with a maximum latitude of 55.98° and a minimum of -41.33° — confirming that while the dataset is global in principle, the overwhelming majority of restaurants cluster around the Indian subcontinent (latitude 25–29°N, longitude 77°E), with smaller clusters in North America, the UK, Southeast Asia, and Australia/New Zealand.

Insight: The data confirms a strong geographic concentration in India (specifically the Delhi NCR region), with the dataset functioning more as an India-centric restaurant directory with a long tail of international listings rather than a truly globally balanced sample.

4.4 Task 4 — Restaurant Chains

Restaurants appearing more than once under the same name were treated as chains. Their branch counts, average ratings, and total customer votes were compared.

Restaurant Chain	Number of Branches
Cafe Coffee Day	83
Domino's Pizza	79
Subway	63
Green Chick Chop	51
McDonald's	48

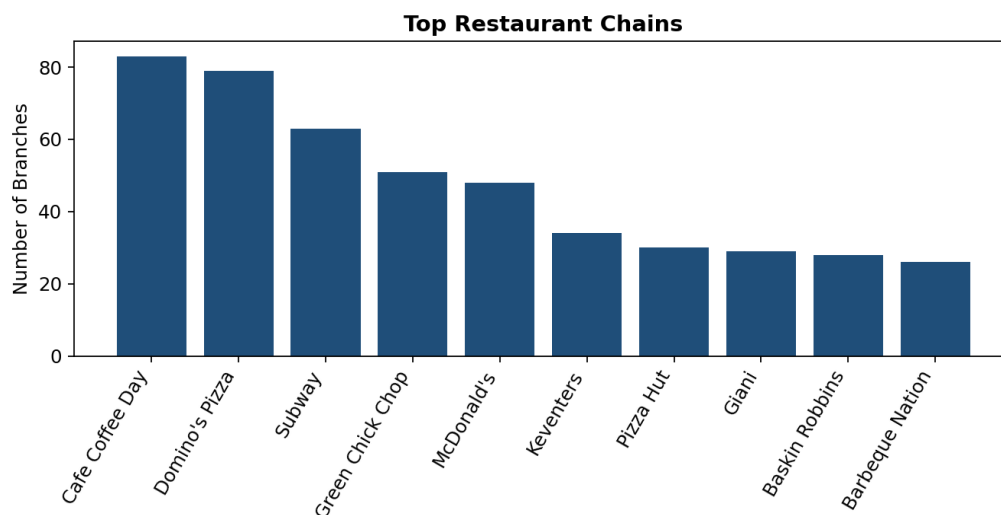


Figure 7: Top restaurant chains by number of branches

Notably, the chains with the most branches (Cafe Coffee Day, Domino's Pizza, Subway) are not the same as the highest-rated chains. The top-rated chains by average rating were Talaga Sampireun (4.90), AB's Absolute Barbecues (4.85), and Silantro Fil-Mex (4.85) — all relatively small, regional chains. Meanwhile, the chains with the highest total customer votes were Barbeque Nation (28,142 votes), AB's – Absolute Barbecues (13,400), and Big Chill (10,853).

Insight: Branch count, average rating, and popularity (vote volume) are three distinct and largely independent dimensions of success. Mass-market chains (Cafe Coffee Day, Domino's, Subway, McDonald's) win on footprint and convenience, while specialty/experience-driven chains (Barbeque Nation, AB's Absolute Barbecues) win on customer engagement and loyalty, generating disproportionately high votes and ratings relative to their smaller branch networks.

5. Level 3 Analysis

Level 3 extends the analysis further into engagement metrics and service-feature pricing patterns, examining how customer voting behaviour relates to ratings, and how price positioning influences the availability of online delivery and table booking.

5.1 Task 1 — Restaurant Name and Reviews

This task required extracting the most frequently used positive and negative keywords from restaurant review text to identify common praise and complaint themes. The supplied dataset does not include a review text column — it contains structured restaurant metadata (cuisine, location, pricing, ratings, votes) but no free-text customer reviews. As a result, keyword extraction and sentiment-based theme analysis could not be performed on this dataset. This was confirmed programmatically and noted directly in the analysis rather than fabricated from unavailable data.

5.2 Task 2 — Votes Analysis

The restaurant with the highest and lowest number of votes was identified, and the correlation between votes and aggregate rating was calculated.

Metric	Restaurant Name	Votes
Highest Votes	Toit	10,934
Lowest Votes	Cantinho da Gula	0

Correlation (Votes vs. Aggregate Rating): 0.314

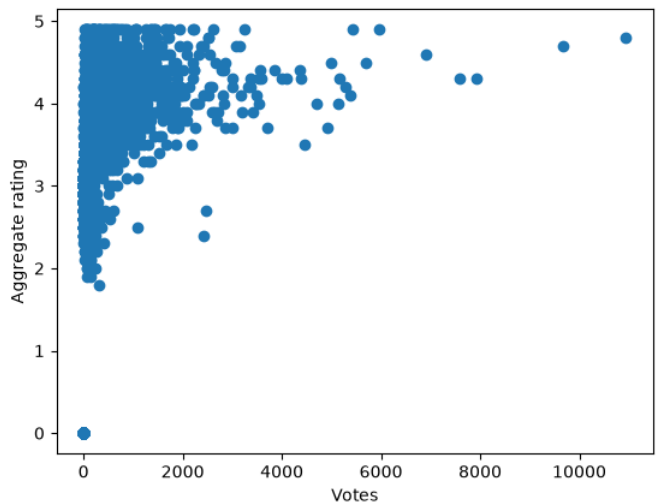


Figure 8: Relationship between number of votes and aggregate rating

Insight: The correlation of 0.31 indicates a weak-to-moderate positive relationship between vote count and rating — highly-voted restaurants tend to skew toward higher ratings, but the relationship is far from strict. The scatter plot shows that restaurants with very few votes span the full rating range (including the 0-rating cluster of unrated restaurants), while restaurants with a high vote count (1,000+) are consistently rated above 3.5. This suggests vote volume acts more as a proxy for restaurant visibility and popularity than as a direct driver of quality perception, though highly popular restaurants rarely sustain a poor rating at scale.

5.3 Task 3 — Price Range vs. Service Availability

The relationship between price range and the availability of two key services — online delivery and table booking — was examined using normalized cross-tabulation.



Figure 9: Online delivery availability by price range

Insight: Online delivery availability decreases sharply as price range increases. Budget restaurants (Price Range 1) and mid-range restaurants (Price Range 2) are far more likely to offer online delivery (roughly 16% and 41% respectively) compared to premium restaurants (Price Range 4), where online delivery drops to just 9%. This suggests that delivery-oriented service models are concentrated in the affordable dining segment, while premium restaurants favour an in-venue dining experience over delivery.

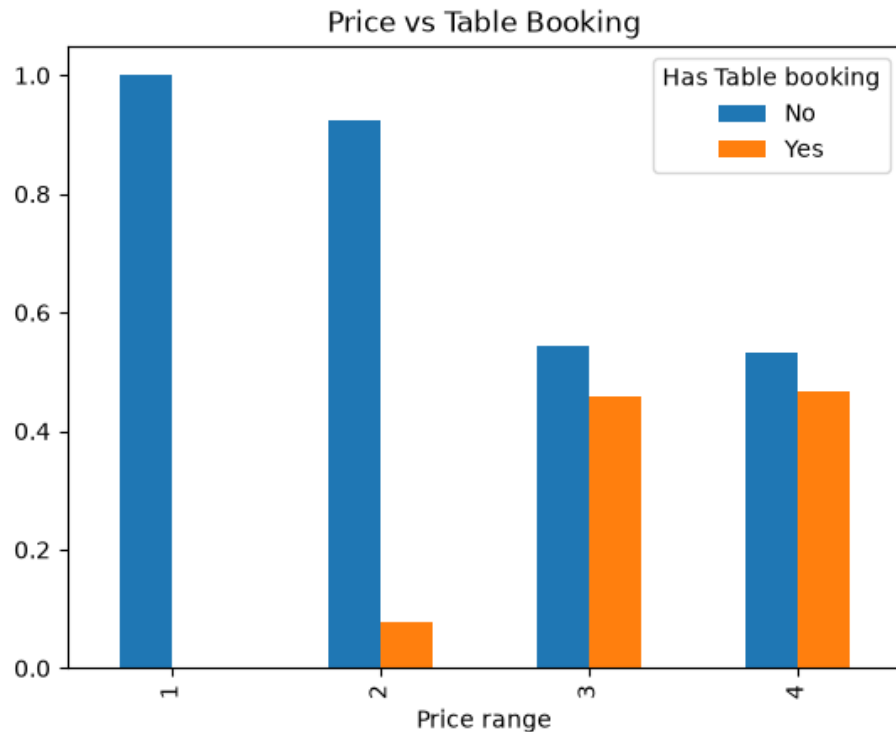


Figure 10: Table booking availability by price range

Insight: Table booking shows the opposite pattern — it is almost entirely absent in the cheapest price tier (0% at Price Range 1) and rises steadily with price, reaching roughly 46–47% at Price Ranges 3 and 4. This is intuitive: table booking is a feature associated with sit-down, reservation-driven dining experiences, which are far more common at higher price points, while budget and fast-casual restaurants (Price Range 1–2) rarely require or offer reservations.

Taken together, these two patterns reveal a clear service-model split by price tier: budget restaurants compete on delivery convenience, while premium restaurants compete on reservation-based, in-venue experience.

6. Key Findings and Business Insights

- North Indian cuisine is the most dominant category, present in over 41% of restaurants, reflecting the dataset's strong representation of North Indian markets.
- New Delhi accounts for the largest share of restaurants by far (5,473 of 9,551), making the dataset effectively NCR-centric despite covering multiple countries.
- Smaller markets (e.g., Inner City, Quezon City) achieve higher average ratings than high-volume markets, suggesting quality is not driven by market saturation.
- Budget-friendly restaurants (Price Range 1–2) make up nearly 80% of all listings, indicating the dataset skews toward everyday, mass-market dining rather than fine dining.
- Online delivery availability is associated with a meaningfully higher average rating (3.25 vs. 2.47), highlighting a strong link between digital service capability and customer satisfaction.

- Ratings cluster around the 3–4 “Good” range for restaurants that have been rated at all, while a substantial portion of listings remain unrated.
- Niche, focused cuisine combinations achieve the highest average ratings, while broad/generalist cuisines win on volume — a quality vs. reach trade-off.
- Restaurant chain success is multi-dimensional: footprint (branch count), quality (rating), and popularity (votes) do not necessarily move together.
- Vote count and rating show only a weak-to-moderate positive correlation (0.31) — high vote counts are associated with consistently decent ratings, but vote volume reflects popularity more than quality.
- Service availability splits cleanly by price tier: budget restaurants favour online delivery, while premium restaurants favour table booking — reflecting two distinct, price-driven service models.

7. Conclusion

This project applied a structured, three-level exploratory data analysis approach to the Zomato Restaurant Dataset, covering cuisine trends, city-level performance, pricing behaviour, service-feature impact, rating distributions, geographic spread, restaurant chain dynamics, customer engagement patterns, and price-driven service-model differences. The analysis demonstrates practical proficiency in Python-based data analysis using pandas for data wrangling and aggregation, and matplotlib/seaborn/folium for static and geospatial visualization.

The findings highlight several actionable insights for restaurant businesses and platform operators alike: the strong value of online delivery in driving customer satisfaction, the trade-off between cuisine specialisation and market reach, the distinction between scale-driven and loyalty-driven success for restaurant chains, and the clear price-tier split between delivery-oriented and reservation-oriented service models. These insights showcase how foundational data analytics techniques can be used to extract meaningful, decision-relevant patterns from real-world business data.

7.1 Skills Demonstrated

- Data loading, inspection, and quality assessment using pandas.
- Categorical data processing (string splitting, exploding multi-value fields).
- Grouped aggregation and statistical summarisation (groupby, mean, value_counts).
- Correlation analysis to quantify relationships between numeric variables (e.g., votes and ratings).
- Data visualization using matplotlib and seaborn (bar charts, histograms, KDE plots, scatter plots).
- Geospatial visualization using folium for interactive mapping.
- Deriving and articulating business insights from quantitative analysis.